Gradient Descent and Automatic Differentiation

Optimization Methods — Week 3 (80 minutes)

DSBA

Learning objectives

By the end of the lecture, you should be able to

- derive gradient descent and select a stable step size;
- explain Armijo backtracking and practical stopping rules;
- prove convergence under convexity, strong convexity, and the PL condition;
- predict how conditioning affects trajectories and rates;
- compute derivatives with forward and reverse automatic differentiation.

The template of an iterative method

Given x_0, construct

$$d_k = \mathcal{D}(x_k, f, \nabla f, \ldots), \qquad \alpha_k = \mathcal{S}(x_k, d_k), \qquad x_{k+1} = x_k + \alpha_k d_k.$$

Three design questions: direction, step, and termination.

Descent direction

d is a strict descent direction at x if $\nabla f(x)^\top d < 0$. For differentiable f, this implies $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$.

Why the gradient?

Theorem (Steepest local ascent in the Euclidean norm)

If f is differentiable at x and $\nabla f(x) \neq 0$, then

$$\arg \max_{\|d\|_2 \leq 1} \lim_{t \downarrow 0} \frac{f(x+td) - f(x)}{t} = \frac{\nabla f(x)}{\|\nabla f(x)\|_2}.$$

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Geometry depends on the norm

For a norm $\|\cdot\|$, the steepest unit direction solves

$$\max_{\|d\|\le 1} \nabla f(x)^\top d = \|\nabla f(x)\|_* ,$$

where $\|\cdot\|_*$ is the dual norm. Thus "negative gradient" means Euclidean geometry. For $\|d\|_M^2 = d^\top M d$, $M \succ 0$,

$$d_{\text{steep}} \propto -M^{-1}\nabla f(x),$$

foreshadowing preconditioning and natural gradient methods.

Gradient descent with a constant step

Definition (Gradient descent)

For $\alpha > 0$,

$$x_{k+1} = x_k - \alpha \nabla f(x_k).$$

One iteration costs one gradient evaluation. A constant step is predictable and cheap, but it must respect the curvature scale.

The descent lemma

Theorem (Descent lemma)

If ∇f is L-Lipschitz, then

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|_2^2 .$$

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$r'(t) = \nabla f(x + t(y - x))^\top (y - x)$. Add and subtract $\nabla f(x)$ and use Cauchy–Schwarz plus Lipschitzness. The remainder is at most $\int_0^1 Lt \|y - x\|^2 dt = \frac{L}{2} \|y - x\|^2$. □

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Therefore

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Every $0 < \alpha < 2/L$ guarantees strict decrease away from stationarity; $\alpha \leq 1/L$ gives the convenient bound $f(x^+) \leq f(x) - \frac{\alpha}{2} \left\|\nabla f(x)\right\|^2$.

Quadratics reveal the exact stability condition

For $f(x) = \frac{1}{2}x^\top Qx - b^\top x$, $Q \succ 0$, let $e_k = x_k - x^*$:

$$e_{k+1} = (I - \alpha Q)e_k.$$

After diagonalizing $Q = U\Lambda U^\top$, each coordinate satisfies

$$z_{k+1,i} = (1 - \alpha\lambda_i)z_{k,i}.$$

Convergence requires $|1 - \alpha\lambda_i| < 1$ for every i, hence

$$0 < \alpha < \frac{2}{\lambda_{\max}(Q)}.$$

At or beyond the boundary: oscillation or divergence.

Armijo backtracking

Choose $\alpha_0 > 0$, shrinkage $\beta \in (0,1)$, and $c \in (0,1)$. Starting at $\alpha = \alpha_0$, repeat

$$\alpha \leftarrow \beta\alpha \quad \text{until} \quad f(x + \alpha d) \leq f(x) + c\alpha \nabla f(x)^\top d.$$

For gradient descent, the right side is $f(x) - c\alpha \|\nabla f(x)\|^2$. Typical values: $\beta = 1/2$, $c = 10^{-4}$.

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The descent lemma gives $f(x + \alpha d) \leq f(x) - \alpha(1 - L\alpha/2)\,\|\nabla f(x)\|^2$. Armijo holds whenever $1 - L\alpha/2 \geq c$. Thus it holds for every $\alpha \leq 2(1 - c)/L$.

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The descent lemma gives $f(x + \alpha d) \leq f(x) - \alpha(1 - L\alpha/2)\,\|\nabla f(x)\|^2$. Armijo holds whenever $1 - L\alpha/2 \geq c$. Thus it holds for every $\alpha \leq 2(1 - c)/L$. Since $\beta^j \alpha_0 \to 0$, a finite j enters this interval. □

Stopping criteria: use more than one signal

$$\|\nabla f(x_k)\| \le \varepsilon_g \max(1, |f(x_k)|),$$
$$\|x_{k+1} - x_k\| \le \varepsilon_x \max(1, \|x_k\|),$$
$$|f(x_{k+1}) - f(x_k)| \le \varepsilon_f \max(1, |f(x_k)|).$$

Also impose maximum iterations/evaluations and detect non-finite values. Small changes can mean convergence, a tiny step, poor scaling, or floating-point stagnation; a small gradient is usually the strongest first-order certificate.

Convex rate: $O(1/k)$

Theorem

Let f be convex and L-smooth, and let x^* be a minimizer. GD with $\alpha = 1/L$ satisfies

$$f(x_k) - f(x^*) \leq \frac{L \|x_0 - x^*\|^2}{2k}.$$

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Strong convexity implies a linear rate

Theorem

If f is μ-strongly convex and L-smooth, GD with $\alpha = 1/L$ obeys

$$f(x_k) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^*).$$

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The PL condition is enough

Definition (Polyak–Łojasiewicz condition)

f satisfies PL with $\mu > 0$ if

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^*) \quad \text{for all } x.$$

Theorem

For an L-smooth PL function, GD with $\alpha = 1/L$ has the same function-value rate $(1 - \mu/L)^k$.

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Use $f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$. Apply PL to the subtracted term. Subtract f^* and iterate. No convexity was used. □

Conditioning controls speed

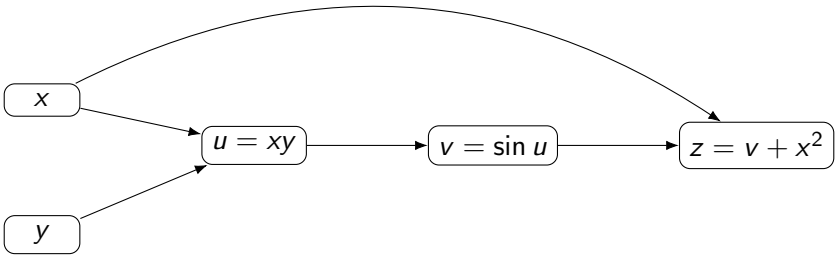
For a quadratic, $\mu = \lambda_{\min}(Q)$, $L = \lambda_{\max}(Q)$, and

$$\kappa = \frac{L}{\mu}.$$

With $\alpha = 1/L$, the contraction factor is $1 - 1/\kappa$; about $O(\kappa \log(1/\varepsilon))$ iterations are needed. The optimal constant step for a quadratic is $2/(L + \mu)$, with factor $(\kappa - 1)/(\kappa + 1)$. Long narrow contours cause zig-zagging; rescaling or preconditioning changes the geometry.

A computation is a directed acyclic graph

For $u = xy$, $v = \sin u$, $z = v + x^2$:



Nodes store values; edges encode dependencies; local derivative rules compose by the chain rule.

Forward mode: propagate tangents

For a direction $\dot{x}$, each node carries $(v, \dot{v})$ with

$$\dot{v} = J_v \dot{x}.$$

Example: $u = xy$, $v = \sin u$, $z = v + x^2$:

$$\begin{aligned} \dot{u} &= y\dot{x} + x\dot{y}, \\ \dot{v} &= \cos(u)\dot{u}, \\ \dot{z} &= \dot{v} + 2x\dot{x}. \end{aligned}$$

One sweep computes a Jacobian–vector product $J\dot{x}$. Efficient when the number of input directions is small.

Reverse mode: propagate adjoints

For scalar output z, define $\bar{v} = \partial z / \partial v$. Traverse nodes backward:

$$\bar{u} \mathrel{+}= \bar{v} \frac{\partial v}{\partial u}.$$

Initialize $\bar{z} = 1$. In the example,

$$\bar{v} = 1, \quad \bar{u} = \cos u, \quad \bar{x} = y \cos u + 2x, \quad \bar{y} = x \cos u.$$

One reverse sweep computes a vector–Jacobian product; for scalar loss and many parameters, this is backpropagation.

Why gradients must accumulate

If a value feeds several downstream operations, the multivariable chain rule adds all paths:

$$z = x^2 + x^3, \qquad \frac{dz}{dx} = 2x + 3x^2.$$

Thus reverse-mode implementations use

$$x.\mathrm{grad} \mathrel{+}= \text{local contribution},$$

not assignment. Across training batches, frameworks also accumulate parameter gradients unless they are explicitly reset.

Backpropagation through a model

For logistic regression,

$$s_i = w^\top x_i + b, \quad p_i = \sigma(s_i), \quad \ell_i = -y_i \log p_i - (1 - y_i) \log(1 - p_i).$$

Reverse mode reuses stored forward values and yields

$$\frac{\partial \ell_i}{\partial s_i} = p_i - y_i, quad \nabla_w \ell_i = (p_i - y_i) x_i, quad \frac{\partial \ell_i}{\partial b} = p_i - y_i.$$

For the mean loss, contributions from observations are summed and divided by n.

AD is not symbolic or numerical differentiation

Method	Core idea	Accuracy	Typical issue
Symbolic	manipulate formulas	exact	expression swell
Finite differences	perturb inputs	approximate	cancellation/truncation
Automatic differentiation	chain rule on program	machine precision	memory/control flow

Reverse mode stores intermediates; checkpointing trades recomputation for memory.
Non-smooth primitives require a chosen subgradient convention.

Takeaways

- Smoothness converts a local gradient into a global safe-step guarantee.
- Convexity gives sublinear convergence; strong convexity or PL gives linear convergence.
- The condition number predicts the practical difficulty of first-order optimization.
- Backtracking adapts to unknown local curvature.
- Reverse-mode AD is the chain rule organized for one scalar loss and many parameters.

Suggested reading: Nocedal and Wright, *Numerical Optimization*, Chs. 2–3; Boyd and Vandenberghe, *Convex Optimization*, Ch. 9; Baydin et al., "Automatic Differentiation in Machine Learning: a Survey."